

§14.3.

Partial derivatives of $f(x, y)$

$$\frac{\partial f}{\partial x}(x_0, y_0) = \lim_{h \rightarrow 0} \frac{f(x_0+h, y_0) - f(x_0, y_0)}{h}$$

$$\frac{\partial f}{\partial y}(x_0, y_0) = \lim_{h \rightarrow 0} \frac{f(x_0, y_0+h) - f(x_0, y_0)}{h}$$

other notations: $f_x(x_0, y_0)$, $f_y(x_0, y_0)$, $\left. \frac{\partial f}{\partial x} \right|_{(x_0, y_0)}$, $\left. \frac{\partial f}{\partial y} \right|_{(x_0, y_0)}$

Implicit differentiation (see §14.4)

second-order partial derivatives

$$\begin{matrix} f_{xx}, & f_{yy}, & f_{xy}, & f_{yx} \\ \frac{\partial^2 f}{\partial x^2}, & \frac{\partial^2 f}{\partial y^2}, & \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) \end{matrix}$$

f_{xy} & f_{yx} are continuous at $(x_0, y_0) \Rightarrow f_{xy}(x_0, y_0) = f_{yx}(x_0, y_0)$

§14.4.

Chain rule for functions of one independent variable

i) $W = f(x, y)$, $x = x(t)$, $y = y(t)$

$$\Rightarrow \frac{dW}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$$

ii) $W = f(x, y, z)$, $x = x(t)$, $y = y(t)$, $z = z(t)$

$$\Rightarrow \frac{dW}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt}$$

Chain rule for functions of two independent variables

i) $W = f(x, y)$, $x = g(r, s)$, $y = h(r, s)$

$$\Rightarrow \begin{cases} \frac{\partial W}{\partial r} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial r} \\ \frac{\partial W}{\partial s} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial s} \end{cases}$$

$$ii) W = f(x, y, z), \quad x = g(r, s), \quad y = h(r, s), \quad z = k(r, s)$$

$$\Rightarrow \begin{cases} \frac{\partial W}{\partial r} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial r} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial r} \\ \frac{\partial W}{\partial s} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial s} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial s} \end{cases}$$

Implicit differentiation

$$i) F(x, y) = 0 \Rightarrow \frac{dy}{dx} = -\frac{F_x}{F_y}$$

$\Rightarrow y$ is implicitly defined as a function of x

$$ii) F(x, y, z) = 0 \rightarrow z \text{ is implicitly defined as a function of } x \text{ and } y.$$

$$\Rightarrow \begin{cases} \frac{\partial z}{\partial x} = -\frac{F_x}{F_z} \\ \frac{\partial z}{\partial y} = -\frac{F_y}{F_z} \end{cases}$$

§14.5

directional derivative, $f(x, y)$, $P_0(x_0, y_0)$, unit vector $u = \langle u_1, u_2 \rangle$

$$(D_u f)_{P_0} = (\nabla f)_{P_0} \cdot u = \langle f_x(x_0, y_0), f_y(x_0, y_0) \rangle \cdot \langle u_1, u_2 \rangle$$

$$\hookrightarrow \text{Gradient vector of } f: \nabla f = f_x i + f_y j$$

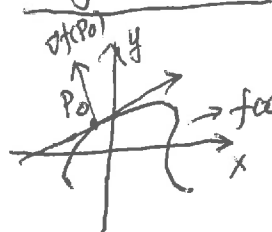
Similarly, for $f(x, y, z)$, $P_0(x_0, y_0, z_0)$, $u = \langle u_1, u_2, u_3 \rangle$

$$(D_u f)_{P_0} = \langle f_x(x_0, y_0, z_0), f_y(x_0, y_0, z_0), f_z(x_0, y_0, z_0) \rangle \cdot \langle u_1, u_2, u_3 \rangle$$

Remark: $D_u f = \nabla f \cdot u = |\nabla f| \cos \theta$ where θ is the angle between u and ∇f .

Tangent line to level curve at a point

$$f(x, y) = k, \quad P_0(x_0, y_0)$$



$$\Rightarrow f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) = 0$$

Rules for gradients ∇f

$$\nabla(f \pm g) = \nabla f \pm \nabla g, \quad \nabla(\alpha f) = \alpha \nabla f$$

$$\nabla(fg) = f\nabla g + g\nabla f, \quad \nabla\left(\frac{f}{g}\right) = \frac{g\nabla f - f\nabla g}{g^2}$$

Chain rule for paths

$\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle \rightarrow$ a smooth curve $\mathcal{C}$

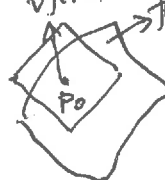
$W = f(\mathbf{r}(t)) \rightarrow$ a scalar function evaluated along $\mathcal{C}$

$$\text{then } \frac{d}{dt} f(\mathbf{r}(t)) = \nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t)$$

§14.6

Tangent plane to the level surface at a point

$\nabla f(P_0) \perp \{ (x, y, z) \in \mathbb{R}^3 : f(x, y, z) = C, P_0(x_0, y_0, z_0) \}$


$$\Rightarrow f_x(P_0)(x - x_0) + f_y(P_0)(y - y_0) + f_z(P_0)(z - z_0) = 0$$

Linearization of a function $f(x, y)$ at a point $P_0(x_0, y_0)$

$$f(x, y) \approx L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

$\downarrow$ standard linear approximation of f at the neighborhood of (x_0, y_0) .

$$(x_0, y_0) \rightarrow (x_0 + \overset{dx}{\Delta x}, y_0 + \overset{dy}{\Delta y})$$

total differential of f $\rightarrow$ differentials of x, y .

$$df = \Delta f \approx \Delta L = f_x(x_0, y_0)dx + f_y(x_0, y_0)dy$$

Similarly, $f(x, y, z), P_0(x_0, y_0, z_0)$

$$L(x, y, z) = f(P_0) + f_x(P_0)(x - x_0) + f_y(P_0)(y - y_0) + f_z(P_0)(z - z_0)$$

$$df = f_x(P_0)dx + f_y(P_0)dy + f_z(P_0)dz$$

§14.7.

Given $f(x, y)$ defined on R containing (a, b)

$f(a, b)$ local maximum value : $f(a, b) \geq f(x, y)$ for $(x, y) \in R$ in a neighborhood of (a, b)

local minimum value : $f(a, b) \leq f(x, y)$ for $(x, y) \in R$ in a neighborhood of (a, b)

interior points, boundary points

First derivative test for local extreme values

$f(x, y)$ has a local extreme value at an interior point (a, b)

$$\Rightarrow f_x(a, b) = f_y(a, b) = 0.$$

Critical points, saddle points

- $\hookrightarrow$ { i) If interior point & $f_x = f_y = 0$
ii) If interior point & first derivatives don't exist,

second derivative test for local extreme values

Assume $f_x(a, b) = f_y(a, b) = 0$. $D = D(a, b) = f_{xx}(a, b)f_{yy}(a, b) - (f_{xy}(a, b))^2$.

i) $f_{xx}(a, b) < 0$, $D(a, b) > 0 \Rightarrow f(a, b)$ is a local maximum.

ii) $f_{xx}(a, b) > 0$, $D(a, b) > 0 \Rightarrow f(a, b)$ is a local minimum.

iii) $D(a, b) < 0 \Rightarrow (a, b)$ is a saddle point of f .

iv) The test is inconclusive at (a, b) if $D(a, b) = 0$.

§14.8.

Method of Lagrange multipliers

max/min $f(x, y, z)$

st. $g(x, y, z) = k$

$\Leftrightarrow$

$$\begin{cases} \nabla f(x, y, z) = \lambda \nabla g(x, y, z) \\ g(x, y, z) = k \end{cases}$$

$\nearrow$ Lagrange multiplier

$$\begin{aligned} \max/\min f(x, y, z) \\ \text{s.t. } \begin{cases} g(x, y, z) = k \\ h(x, y, z) = c \end{cases} & \Leftrightarrow \begin{cases} \nabla f(x, y, z) = \lambda \nabla g(x, y, z) + \mu \nabla h(x, y, z) \\ g(x, y, z) = k \\ h(x, y, z) = c \end{cases} \end{aligned}$$

Lagrange multipliers

Chapter 15.

§ 15.1.

$f(x, y)$, $R = [a, b] \times [c, d] \leftarrow \text{rectangle}$

Double integral

$$\iint_R f(x, y) dA = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n f(x_i, y_j) \Delta A$$

where subrectangles $R_{ij} = [x_i, x_{i+1}] \times [y_j, y_{j+1}]$
 $\Delta A = \Delta x \Delta y$

Fubini's Theorem (First form)

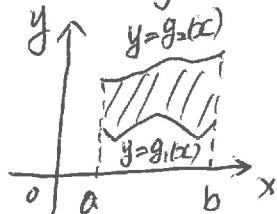
f is continuous. Then

$$\iint_R f(x, y) dA = \int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy$$

§ 15.2.

Types of planar regions, R

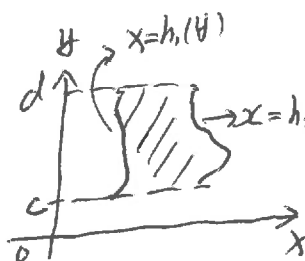
TYPE I:



$$R = \{(x, y) \mid a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$$

Then
$$\iint_R f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx$$

TYPE II:



$$R = \{(x, y) \mid c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$$

Then
$$\iint_R f(x, y) dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy$$

Properties of double integrals

$$1. \iint_R \alpha f(x, y) dA = \alpha \iint_R f(x, y) dA$$

$$2. \iint_R (f(x, y) \pm g(x, y)) dA = \iint_R f(x, y) dA \pm \iint_R g(x, y) dA$$

$$3. (a) \iint_R f(x, y) dA \geq 0 \text{ if } f(x, y) \geq 0 \text{ on } R$$

$$(b) \iint_R f(x, y) dA \geq \iint_R g(x, y) dA \text{ if } f(x, y) \geq g(x, y) \text{ on } R.$$

$$4. \text{ If } R = R_1 \cup R_2 \text{ and } R_1 \cap R_2 = \emptyset, \text{ then}$$

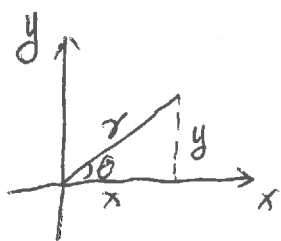
$$\iint_R f(x, y) dA = \iint_{R_1} f(x, y) dA + \iint_{R_2} f(x, y) dA$$

§ 15.3.

$$\text{Area}(R) = \iint_R dA$$

$$\text{Average of } f \text{ over } R = \frac{1}{\text{Area}(R)} \iint_R f dA.$$

§ 15.4.

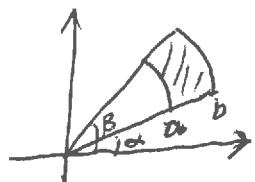


Polar coordinates (r, θ) .

$$r = \sqrt{x^2 + y^2}, \quad \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$x = r \cos \theta, y = r \sin \theta$ for cartesian coordinates.

A planar rectangle

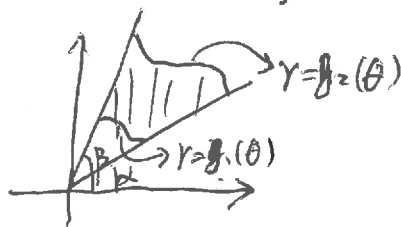


$$R = \{(r, \theta) \mid a \leq r \leq b, \alpha \leq \theta \leq \beta\}$$

$$\iint_R f(x, y) dA = \int_{\alpha}^{\beta} \int_a^b f(r \cos \theta, r \sin \theta) r dr d\theta$$

A polar region R given by

$$R = \{(r, \theta) / \alpha \leq \theta \leq \beta, g_1(\theta) \leq r \leq g_2(\theta)\}$$



$$\iint_R f(x, y) dA = \int_{\alpha}^{\beta} \int_{g_1(\theta)}^{g_2(\theta)} f(r \cos \theta, r \sin \theta) \underline{r dr d\theta}$$